Does the Conformal Guarantee Travel? Measuring, Auditing and Repairing Coverage Loss in Cross-Dataset Trajectory Prediction

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***Abstract*—A conformal prediction region promises to contain a road user's true future path a fixed fraction of the time. We ask whether that promise survives when a trajectory forecaster is calibrated on one driving dataset and deployed, without new labels, on another. Using forecasters calibrated on Argoverse 2 or nuScenes and evaluated on each other and on Waymo Open Motion, we measure coverage losses of 8 to 23 percentage points at a 90 percent nominal level. The losses are confirmed on models that match published accuracy and were registered in advance for the third dataset. The most accurate model loses the most coverage. Reweighting the calibration data by an estimated covariate-shift likelihood ratio makes the loss worse, and we show that no label-free method can repair an unrestricted shift. The loss concentrates in turns, most of all in left-hand traffic, and measured input differences between the datasets explain only about a quarter of it. Labelled target data can repair it, but labels arrive one scene at a time, and a guarantee that treats agents as independent fails when scenes contain many agents. A scene-level Learn-then-Test rule with a betting test restores the guarantee at every labelling budget we tested, and returns a region within twice the oracle size from about 150 to 200 labelled scenes. Conformal guarantees for trajectory prediction should therefore not be assumed to transfer across datasets without an audit and a scene-level recalibration.**

***Index Terms*—Conformal prediction, trajectory prediction, distribution shift, autonomous driving.**

AUTONOMOUS vehicles plan around forecasts of what nearby road users will do. A forecast is only as useful as the statement it makes about its own reliability. A planner that knows a predicted region contains the true future with probability at least $1-\alpha$ can keep clear of that region and of nothing larger. Conformal prediction (CP) supplies such a statement for any black-box predictor, with a finite-sample guarantee that needs only exchangeability of calibration and test data [1], [2], and it is now used for trajectory prediction and safe planning [3]–[6].

Deployment breaks exchangeability. A forecaster is calibrated on the data at hand, with one set of cities, one sensor suite and one annotation pipeline, and is then driven somewhere else. Trajectory predictors lose accuracy when moved between datasets [7], [8], and model calibration degrades under dataset shift [9]. What is not known is what happens to the *guarantee itself* when a conformally calibrated forecaster is moved, without new labels, to a different real driving dataset: whether the stated $1-\alpha$ survives, how badly it fails, and whether the standard remedies repair it. Weighted CP needs the likelihood ratio [10], distance-based bounds need the size of the shift [11], [12], and online methods need a stream of labels [13], [14]. None settles the static, label-free, dataset-to-dataset case that Fig. 1 illustrates.

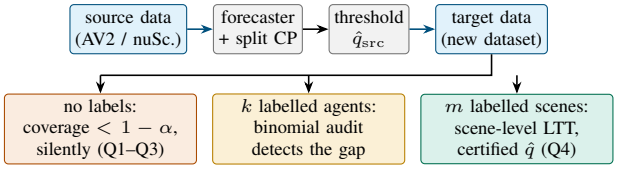


Fig. 1. The transfer problem and the paper's questions. A threshold calibrated on one dataset is deployed on another. Without labels the coverage loss is invisible; k labelled agents detect it; m labelled scenes repair it.

Contributions: (i) A direct measurement of zero-label conformal coverage transfer between real driving datasets, in both directions between Argoverse 2 and nuScenes and, registered before it was run, on Waymo Open Motion, which to our knowledge no prior work has quantified at this scale. Coverage falls by up to 23.4 points at a nominal 90%, on models that match published accuracy, and the most accurate model loses the most. (ii) A diagnosis. Covariate reweighting makes the loss worse, a non-identifiability result (Proposition 1) shows why label-free repair is limited in principle, and controlled input perturbations localise the loss to turns in left-hand traffic. (iii) A repair with a finite-sample, training-conditional guarantee when labels arrive scene by scene (Propositions 2 and 3). It is valid at every labelling budget we tested and gives a usable region from about 150 labelled scenes, while the natural agent-level guarantee fails when scenes contain many agents. (iv) Code, registered hypotheses and an append-only experiment log, including a defect found and corrected before submission.

I. Related Work

a) Conformal prediction for trajectories.: Split CP turns any predictor into one with marginal coverage guarantees [1], [2] and underlies conformal safe planning around predicted agents [3]. Multi-step and multi-modal trajectories have been handled with copula-based intervals [4], scenario-aware intervals in a Frenet frame on nuPlan with a learned reliability discriminator [15], and multi-view calibration that fuses

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vehicle- and infrastructure-side observations before conformal calibration [16]. A 2025 survey of the field catalogues these directions and the open gaps around uncertainty and generalisation [17]; concurrent architecture work targets multimodal mode decoding [18] and a shared encoder across simulation, prediction and planning [19], and a controlled dataset-design study on a new benchmark finds that feature engineering matters less than which data the model sees [8] — a finding our Sec. V-C echoes from the calibration side. Calibrating the predictive *distribution* itself, rather than only the output region, has also been proposed [20]. Locally weighted scores that divide the residual by an estimated spread [21] are the classical route to adaptive regions; our normalised score uses the forecaster's own predicted scale in that role, an LLM-based safe-navigation system uses a related idea in token space [22].

b) Safe planning and robot safety with conformal guarantees.: Beyond trajectory regions, conformal prediction has been used to certify learned perception for navigation [23], to adapt a crowd-navigation policy's safety margin to distribution shift [24], to learn safety regions from sparse human feedback [25], to certify learned dynamical-system motion primitives [26], and more broadly to connect conformal prediction to formal verification and control of learning-enabled autonomous systems [6]. Closest to our label-free-versus-labelled distinction, [27] transfers a safety guarantee across policy updates analytically rather than by relabelling, and [28] treats the distribution shift the ego agent itself induces in interacting agents. All of these assume the calibration and deployment *domain* match; we study what happens when the domain itself changes.

c) Conformal prediction under shift.: Weighted CP handles covariate shift given the likelihood ratio [10], and the coverage loss under general shift can be bounded by a distance between calibration and test distributions [11]. Recent work replaces that bound with an explicit ambiguity-set radius under Lévy–Prokhorov perturbations [12], regularises the representation with a Wasserstein penalty to shrink the achievable coverage gap [29], or uses optimal transport with unlabelled target data to both estimate and mitigate the loss in coverage [30]; a parallel line gives sample-complexity lower bounds for when a selective predictor can certify a risk floor under covariate shift at all [31], a negative result in the same spirit as our Proposition 1. Online methods adapt the threshold from realised errors: adaptive conformal inference [13], conformal PID control [14], its use for trajectories in CUQDS [5], and adaptive nonconformity scores with a replay buffer [32]. More recent online variants retroactively revise past predictions for faster adaptation [33], cast the threshold update as a universal-portfolio game [34], or give training-conditional (not just time-averaged) regret bounds under drift [35]. The closest trajectory-specific instance of this family is CUQDS [5], which pairs a learned Gaussian-process uncertainty module with online conformal PID control; like the other online methods above, it needs a running stream of ground-truth labels at deployment time, so it answers a different question from ours and is not a baseline we can run in the zero-label or fixed-labelled-batch settings this paper studies. Our ACI comparison (Sec. V-E) covers the same methodological category — adapt the threshold from realised errors — without requiring CUQDS's architecture-specific uncertainty module. Other work conditions on an observable nuisance parameter and generates synthetic data with a diffusion model [36] or treats the shift the ego vehicle itself induces in other agents [28] (both preprints). We study a different point in this space: no feedback, no generative model, and three real driving datasets.

d) Grouped, hierarchical and conditional coverage.: When observations come in groups, exchangeability across observations fails, and hierarchical conformal methods restore validity by pooling or subsampling at the group level [37], [38]; recent extensions handle hierarchically structured outputs with a reconciliation step [39], weight groups directly when the shift is known to act at the group level [40], carry constrained-complexity representations into hierarchical classification [41], or let a test group borrow structure from a donor group when only partial test-group data is available [42]. A broader conditional-coverage literature asks for validity on subpopulations or individual test points rather than only on average: a unifying decomposition of conditional miscoverage [43], learned reliability estimators that predict where coverage will fail [44], [45], group-conditional coverage over overlapping, fractional group memberships [46], an audit showing nominal marginal coverage can still silently fail a specific clinical subgroup [47], and conditional risk control for segmentation via a weighted-quantile connection to risk control [48]. Risk-controlling procedures [49] and Learn-then-Test [50] give finite-sample control of a monotone risk by testing a fixed ordered set of thresholds, and betting-based p-values [51] make such tests far tighter than Hoeffding-type bounds when losses are small. We combine the two ideas for a setting those works do not address: dataset shift in trajectory prediction, where the natural labelling unit is the *scene* and the risk to be controlled is the scene-averaged miscoverage of a fixed, source-calibrated forecaster. Valid p-values for bounded losses need care [52]; our own implementation needed a corrected bet cap (Sec. III-E).

e) Risk control and betting, more generally.: Beyond Learn-then-Test, risk control has been extended to adaptive, data-dependent tradeoffs among several risks at once [53], to a semi-supervised setting that uses unlabelled data to tune the risk-control hyperparameter without extra labels [54], and to post-processing an existing decision policy only where a risk budget is violated, rather than replacing it outright [55]. E-values have been proposed as a more flexible alternative to the p-values we use, enabling anytime-valid and fixed-size variants [56]; weighted-conformal test martingales extend betting-style monitoring to post-deployment shift detection that distinguishes benign covariate shift from harmful shift [57]; and [58] revisits the randomness-theoretic foundations that justify treating a conformal predictor as valid for i.i.d., not only exchangeable, data. We use the same betting machinery for a different purpose: certifying a *scene-averaged* risk from whole labelled scenes.

f) Cross-dataset trajectory prediction and perception under domain shift.: UniTraj harmonises several driving datasets and reports that accuracy drops when models are transferred

between them [7]; that study is about point accuracy, not about the validity of an uncertainty guarantee. Beyond trajectories, [9] showed that calibration of deep models degrades under dataset shift; our question is the conformal analogue for structured, multi-modal forecasts. The same cross-dataset gap is documented on the perception side of the autonomous-driving stack: visual localisation uncertainty is miscalibrated under novel weather and lighting [59], collaborative-perception benchmarks struggle to generalise across sensor rigs and sites [60], cross-sensor pretraining is used specifically to close a domain gap that would otherwise need target-domain labels [61], [62], and a broader uncertainty-aware perception survey argues for explicit, communicable uncertainty rather than a single calibrated score [63]. None of this literature asks whether a *conformal* guarantee specifically, rather than calibration quality in general, survives the same transfer; that is the question this paper measures, diagnoses and partially repairs.

II. Problem Setting

A. Conformal regions for multi-modal trajectories

A forecaster f maps a scene x to K candidate futures $\hat{y}_1, \dots, \hat{y}_K \in \mathbb{R}^{T\times 2}$ for a target agent whose true future is y. We use the nonconformity score

$$s(x, y) = \min_{k\le K} \max_{t\le T} \|\hat{y}_{k,t}(x) - y_t\|_2, \tag{1}$$

so that $s \le q$ iff y lies inside the union of the K tube-shaped balls of radius q around the candidates. Given n calibration scenes from a *source* dataset, split CP sets $\hat{q} =$ the $\lceil (n+1)(1-\alpha)\rceil$-th smallest score, and the region $C_{\hat{q}}(x) = \{y : s(x, y) \le \hat{q}\}$ has $\Pr[y \in C_{\hat{q}}(x)] \ge 1-\alpha$ when the test scene is exchangeable with the calibration scenes [1], [2].

B. The transfer problem

Let P_S, P_T be the source and target scene–future distributions. The *coverage gap* of a source-calibrated threshold on the target is

$$\Delta = (1-\alpha) - \Pr_{P_T}\big[s(x, y) \le \hat{q}\big], \tag{2}$$

positive when the guarantee fails (under-coverage). We ask (Q1) how large Δ is on real data; (Q2) whether covariate-shift reweighting [10] removes it; (Q3) whether anything label-free can; and (Q4) how many labelled target agents are needed to detect it, and how many labelled target scenes to repair it.

C. What label-free repair can and cannot do

Covariate reweighting is valid when $P_T(y \mid x) = P_S(y \mid x)$. If the conditional differs—annotation pipelines, sensor noise, and agent behaviour all differ across datasets—no reweighting on x repairs the gap. More strongly, without restricting the class of shifts nothing label-free can.

Proposition 1 (Non-identifiability of a label-free threshold)**.** *Let S_0 be a nonconformity score with a continuous distribution on the target, and consider the scale-shift family $S = c\,S_0$, $c \ge 1$, whose members are all indistinguishable from unlabeled target inputs. Let $\hat{q}$ be any threshold computed without target labels, so its law does not depend on c. If $\Pr[cS_0 \le \hat{q}] \ge 1-\alpha$ for every $c \ge 1$, then $\Pr[\hat{q} = \infty] \ge 1-\alpha$.*

Proof. For fixed $\hat{q} < \infty$, $\mathbf{1}\{cS_0 \le \hat{q}\} \to 0$ pointwise as $c \to \infty$. By dominated convergence, $\mathbb{E}\Pr[cS_0 \le \hat{q} \mid \hat{q}] \to \Pr[\hat{q} = \infty]$, and the coverage requirement gives $\Pr[\hat{q} = \infty] \ge 1-\alpha$. □

Remark 1. *The proposition does not say label-free methods are useless; it says any guarantee* for all *scale shifts is vacuous, so a useful label-free method must either restrict the shift class (as our normalised score does, Sec. III-A) or be complemented by a label-based audit (Sec. III-B).*

D. Cost of using target labels

If k labelled target agents are used directly, the resulting empirical coverage is Beta-distributed around $1-\alpha$ [64], with standard deviation about $\sqrt{\alpha(1-\alpha)/k}$, so ≈ 1000 labels are needed for ± 2 points at $\alpha = 0.1$. Detecting a shortfall δ with an exact one-sided binomial test at level ℓ and power $1-\beta$ needs

$$k \approx \left(\frac{z_{1-\ell}\sqrt{\alpha(1-\alpha)} + z_{1-\beta}\sqrt{(\alpha+\delta)(1-\alpha-\delta)}}{\delta} \right)^2. \tag{3}$$

For $\delta = 0.034$, $\alpha = 0.1$, $\ell = 0.05$, $\beta = 0.1$ this gives $k \approx 750$.

III. Methods

A. Model-normalised score (label-free)

The forecaster emits, per mode k, a Laplace scale $(b^x_{k,t}, b^y_{k,t})$. Let $u_k(x) = \frac{1}{T}\sum_t \sqrt{(b^x_{k,t})^2 + (b^y_{k,t})^2}$ and use the score $s'(x, y) = \min_k \max_t \|\hat{y}_{k,t} - y_t\| / u_k(x)$. The region is the union of balls of radius $\hat{q}'\,u_k(x)$. The premise is that when a shift makes scenes harder, the forecaster's own scale widens, so the normalised score is less shift-sensitive. Its efficiency is measured by region area $\sum_k \pi(\hat{q}' u_k)^2$ at matched coverage.

B. Coverage audit

Given a source-calibrated $\hat{q}$ and k labelled target agents, an exact one-sided binomial test of H_0: coverage $\ge 1-\alpha$ flags a violation. We report its empirical power and its false-alarm rate on a shift-free split, alongside the theoretical budget of Eq. (3).

C. Recalibration with k target labels

We compare three thresholds computed from k labelled target agents and the n source calibration scores: *direct* (SCP on the k target scores); *pooled* (SCP on the union); and *shrinkage* ($\hat{q} = \lambda\hat{q}_{\text{tgt}} + (1-\lambda)\hat{q}_{\text{src}}$ with $\lambda = k/(k+k_0)$). We report the label budget $k^\star$ at which absolute coverage error falls below 0.02 in at least 90% of random label draws.

D. Certify-or-recalibrate with exchangeable labels

Plain recalibration on k target labels is only *marginally* valid: the coverage of the resulting threshold is $\mathrm{Beta}(\ell, k+1-\ell)$-distributed [64], so it falls below $1-\alpha$ in roughly half of all label draws (Sec. V). A practitioner needs the stronger, *training-conditional* statement [64], [65]: coverage $\geq 1-\alpha$ with probability $\geq 1-\delta$ over the draw of the labels. *Certify-or-recalibrate* (C-or-R) delivers it while keeping the source threshold when it can be certified. With k labelled scores $s_1, \ldots, s_k$ and $M = \#\{s_i > \hat{q}_{\mathrm{src}}\}$:

1) *Certify.* Keep $\hat{q}_{\mathrm{src}}$ iff the one-sided Clopper–Pearson upper confidence bound [66] on its miss probability at level $\delta/2$ is $\leq \alpha$.
2) *Otherwise recalibrate.* Return the ℓ-th smallest labelled score, with ℓ the smallest integer such that $\Pr[\mathrm{Bin}(k, 1-\alpha) \leq \ell - 1] \geq 1 - \delta/2$ (the ℓ-th order statistic has coverage $\mathrm{Beta}(\ell, k+1-\ell)$); if no such $\ell \leq k$ exists return $+\infty$.

Proposition 2 (C-or-R is training-conditionally valid)**.** *If the k target scores are exchangeable with the target scores at test time and continuously distributed, the returned threshold $\hat{q}$ satisfies* $\Pr\left[\Pr_{\textit{test}}(s \leq \hat{q} \mid \textit{labels}) \geq 1-\alpha\right] \geq 1-\delta$.

Proof. Let F be the target score CDF. The event "$\hat{q}$ fails" is contained in (a) the source threshold is certified although $F(\hat{q}_{\mathrm{src}}) < 1-\alpha$, or (b) the branch-2 order statistic has coverage $< 1-\alpha$. (a) has probability $\leq \delta/2$ because the Clopper–Pearson bound is a valid one-sided confidence bound for the miss probability $1 - F(\hat{q}_{\mathrm{src}})$ of the fixed threshold; (b) has probability $\leq \delta/2$ by the definition of ℓ. A union bound gives δ. □

Certification needs slack: a source threshold calibrated at exactly α has miss probability $\approx \alpha$ and its confidence bound cannot fall below α; in our experiments it is (almost) never certified on shifted data, and on unshifted data it is certified only if calibrated at a smaller $\alpha_{\mathrm{src}} < \alpha$.

E. Scene-level risk control under clustered labels

Driving datasets label many agents per scene, and agents in one scene share road, traffic and annotator. Labels drawn scene by scene are not exchangeable at the agent level, and Proposition 2 then fails (Sec. V). We therefore make the *scene* the unit. For a labelled scene j with agent scores $\{s_{ji}\}$ let $L_j(q) = \frac{1}{n_j}\sum_i \mathbf{1}[s_{ji} > q] \in [0,1]$, non-increasing in q, and control the scene-averaged miscoverage $R(q) = \mathbb{E}_j\, L_j(q)$, which weights every *scene* equally regardless of how many agents it contains. This is a different quantity from the agent-pooled miss rate $\frac{1}{\sum_j n_j}\sum_{j,i} \mathbf{1}[s_{ji} > q]$, which weights every agent equally and so is dominated by whichever scenes happen to be largest; the two coincide only when every scene has the same agent count. We control $R(q)$ because it is the quantity the scene-level test (below) can be made valid for under scene-level exchangeability, and because it is the natural risk when scenes, not agents, are the unit a planner reasons about. Given m labelled scenes, scene-level certify-or-recalibrate proceeds as in Sec. III-D but with Learn-then-Test [50] over a *fixed*, label-independent grid $q_1 > q_2 > \cdots > q_G$ (log-spaced around the source threshold, $\hat{q}_{\mathrm{src}} \times [4, 0.5]$, $G = 300$): for each candidate we compute a p-value for $H_q : R(q) > \alpha$ from the m per-scene losses, test candidates from the largest downwards at level $\delta/2$, and return the last q rejected ($+\infty$ if none). The source threshold is tried first at level $\delta/2$ (certification). Two p-values are used: Hoeffding–Bentkus [49], and a betting p-value [51]: with predictable, variance-adaptive bets $\lambda_i \in [0, \frac{1}{2(1-\alpha)}]$ the capital $K = \prod_i \big(1 + \lambda_i(\alpha - L_i)\big)$ stays non-negative for $L_i \in [0,1]$ and is a supermartingale under $\mathbb{E}L \geq \alpha$, so $p = \min(1, 1/\max_t K_t)$ is valid by Ville's inequality. The cap matters: any bet above $1/(1-\alpha)$ lets a single fully missed scene drive the capital negative, which voids the argument.

Proposition 3 (Scene-level guarantee)**.** *If the m labelled scenes are exchangeable draws from the target scene population, and each p-value is valid for its null, the returned threshold satisfies* $\Pr[R(\hat{q}) \leq \alpha] \geq 1-\delta$.

Proof. R is non-increasing, so along the descending grid the false nulls ($R(q_i) \leq \alpha$) form a prefix and the true nulls a suffix beginning at some i_0. The procedure returns a threshold with $R > \alpha$ only if it rejects $H_{q_{i_0}}$ (fixed-sequence testing stops at the first non-rejection), which has probability $\leq \delta/2$ because the grid does not depend on the labels; the certification branch fails with probability $\leq \delta/2$. A union bound gives δ. □

F. Online baseline

Adaptive conformal inference (ACI) [13] updates the working level $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$ after every revealed label, keeping the source calibration set fixed; it needs a stream of labels and offers long-run, not finite-sample, coverage. We report (i) how many streamed labels ACI needs before its running coverage stays within 2 points of nominal, and (ii) the coverage of its threshold frozen after k labels, for $\gamma \in \{0.005, 0.01, 0.05\}$.

G. Controlled input shifts

To locate the source of the loss we perturb one measured AV2–nuScenes difference at a time in AV2 test *inputs* (ground truth untouched) and record the coverage lost by the AV2-calibrated threshold: (i) *2 Hz history*: keep the five 2 Hz keyframes of the 2 s history and interpolate linearly; (ii) *lane dropout*: drop each map polyline with the probability implied by the measured median number of lanes near the target agent (nuScenes 52 vs. AV2 85); (iii) both; (iv) *scene mix*: re-weight AV2 scenes to the nuScenes distribution of seven label-free scene factors with cross-fitted domain-classifier weights, which isolates pure covariate shift. Two further candidates were *measured* and excluded: excess history noise (a quadratic-fit residual estimate is smaller on nuScenes: 0.13 m vs. 0.29 m) and map-point density (equal per scene).

H. Label-free shift monitor

A domain classifier on label-free scene features (native sampling interval, nearby-agent count, map density, nearby lanes, history speed, heading change, curvature) is trained to

separate source and target scenes; its AUC is the monitor. Within a dataset, the same classifier is trained per city against the rest. We test whether the AUC predicts the coverage loss (across cities, by Spearman correlation).

IV. Experimental Setup

a) Data: Argoverse 2 [67], nuScenes [68] and Waymo Open Motion [69] (30 of 150 validation shards), converted with ScenarioNet [70] and loaded through UniTraj [7]. AV2 scenarios are independent 11 s clips with one focal agent; the labelled AV2 validation split is divided by a fixed salted hash into a calibration half and a test half. nuScenes has a mean of 65 agents per 20 s scene (9041 from 138 validation scenes), so its calibration/test split is made by scene (64/74 scenes). Waymo scenarios carry 3.7 agents on average and are split by scenario (4331 calibration, 4318 test).

b) Forecasters: AutoBot-Ego [71] with $K = 6$ modes. The *confirmatory* models were trained after the hypotheses about them were registered, with UniTraj's recipe (effective batch 128, learning rate 7.5×10^{-4} halved at epochs 10–50): **av2_gpu_full** on the full 183k-scene AV2 training set and **ns_gpu_full** on the full nuScenes training split. Each had to reach validation minADE_6 within 15% of UniTraj's published value [7]; they reached 0.854 (gate 0.98, reference 0.85) and 1.232 (gate 1.39, reference 1.21). The final model is the last checkpoint, fixed in advance. Four *reduced-budget* models (training-data subsets or shorter schedules; the three AV2 models reach minADE_6 1.09–1.35) were used to design the analyses and are labelled exploratory throughout.

c) Protocol: $\alpha = \delta = 0.1$. Forward: calibrate on the AV2 calibration half; test on the AV2 test half (same-domain control), nuScenes and Waymo. Reverse: calibrate on the nuScenes calibration scenes; test on nuScenes test scenes, AV2 and Waymo. Label-budget experiments draw k agents or m whole scenes 1000 times from a labelled pool that is disjoint from the evaluation pool. Intervals on multi-agent targets resample scenes.

d) Registration and disclosures: Hypotheses, thresholds and success criteria were written to a version-controlled, timestamped log (released with the code) before the runs they govern; later analyses were added as dated addenda, and outcomes are appended, never edited. Defects found during the work are disclosed there: a cache collision, per-agent instead of per-scene nuScenes splits, and truncated scenario identifiers, none affecting a reported number; and a betting p-value whose bet cap allowed negative capital. That last defect was found in an internal review before submission, a regression test that detects it was added, and every repair result in Sec. V-E was recomputed with the corrected code under a design registered before the rerun.

V. Results

Confirmatory results come first, organised by the four questions of Sec. II. The reduced-budget models on which the analyses were designed are summarised in Sec. V-F.

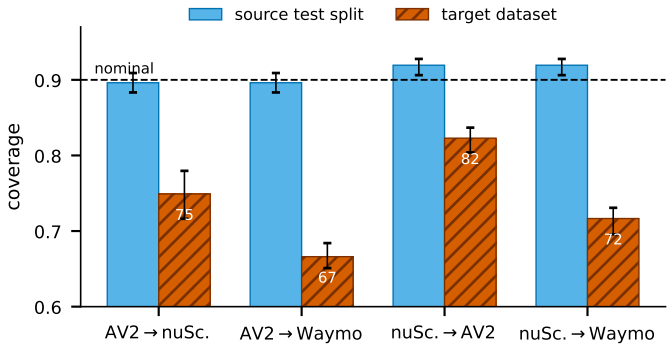


Fig. 2. Coverage of the source-calibrated region ($\alpha = 0.1$) on the source test split and on the target dataset, confirmatory models. Error bars: 95% intervals, resampled by scene where targets have several agents per scene (nuScenes, Waymo).

TABLE I
COVERAGE GAP BY MODEL ($\alpha = 0.10$, POINTS; TARGET NUSCENES FOR FWD., AV2 FOR REV.). CONF.: GATED FULL-DATA MODEL; EXPL.: REDUCED-BUDGET MODEL USED TO DESIGN THE ANALYSES. NORM.: GAP OF THE NORMALISED SCORE. REW.: CHANGE IN GAP FROM COVARIATE REWEIGHTING (POSITIVE = WORSE). AUDIT: REJECT PROBABILITY / FALSE-ALARM % AT $k = 1000$ LABELS.

Model	St.	Dir.	gap	95% CI	norm.	rew.	audit
av2_gpu_full	conf.	fwd	15.1	13.6–16.7	11.0	+1.9	1.00/7
ns_gpu_full	conf.	rev	7.7	6.3–9.6	8.6	+0.1	1.00/0
av2_cpu_v1	expl.	fwd	3.3	2.1–4.5	1.7	+2.0	0.95/2
av2_valsplit_v1	expl.	fwd	8.0	6.4–9.4	7.0	+2.1	1.00/1
av2_cpu_v2	expl.	fwd	10.5	9.4–12.2	7.3	+2.2	1.00/6
ns_cpu_v1	expl.	rev	4.7	3.7–6.2	7.1	-0.5	1.00/19

A. Q1: How large is the gap?

On its own test split each confirmatory model is calibrated as promised: the same-domain gap is 0.4 points for the AV2 model and -1.9 points for the nuScenes model (Fig. 2). On every other dataset it under-covers. Calibrated on AV2, coverage falls by 15.1 points on nuScenes and by 23.4 points on Waymo. Calibrated on nuScenes, it falls by 7.7 points on AV2 and by 18.3 points on Waymo. All four intervals exclude zero by a wide margin, so a Holm correction over the four registered tests changes no conclusion. The Waymo tests were registered before any Waymo prediction was computed. In metres, the AV2-calibrated tube has radius 3.8 m, while 90% coverage needs 8.2 m on nuScenes and 8.9 m on Waymo: the source region is less than half as wide as the target requires.

The nuScenes test agents come from only 138 scenes, so they are strongly dependent. Resampling whole scenes inflates the variance of the coverage estimate by a design effect of 3.9; all intervals on multi-agent targets therefore resample scenes, not agents. The more accurate model has the larger gap. The confirmatory AV2 model, within 0.5% of UniTraj's published accuracy, loses 15.1 points, against 3.3 to 10.5 points for the three reduced-budget AV2 models (Table I). With four models of one architecture family this is an association, not a causal claim.

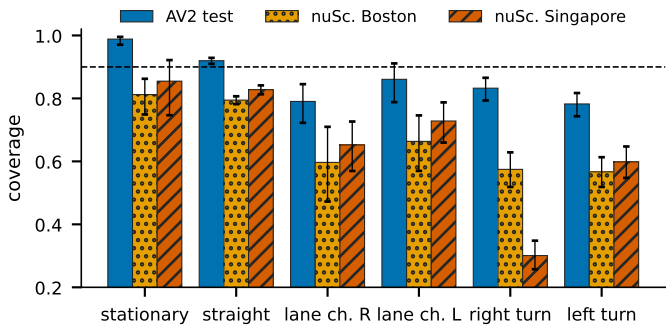


Fig. 3. Coverage by manoeuvre for the confirmatory AV2 model ($\alpha = 0.1$, dashed = nominal). nuScenes is split by city; Singapore drives on the left. Error bars: agent-level Wilson 95% intervals.

TABLE II
WHERE DOES THE LOSS COME FROM? COVERAGE LOST (POINTS; SHARE OF THE REAL GAP G IN PARENTHESES) WHEN ONE MEASURED AV2–NUSCENES DIFFERENCE IS INJECTED INTO AV2 TEST INPUTS: 2 HZ HISTORY, LANE-GRAPH DROPOUT, BOTH, AND RE-WEIGHTING AV2 SCENES TO NUSCENES' SCENE MIX. EXCESS ANNOTATION NOISE AND MAP-POINT DENSITY WERE MEASURED AND FOUND ABSENT (NUSCENES IS SMOOTHER; EQUAL DENSITY), SO THEY ARE NOT INJECTED.

Model	Set	G (pt)	2 Hz	lanes	both	scene mix
av2_cpu_v1	expl.	+3.3	+0.6 (19%)	+1.4 (43%)	+1.8 (53%)	-1.3 (-38%)
av2_valsplit_v1	expl.	+8.0	+0.7 (8%)	+1.4 (17%)	+1.4 (17%)	-1.4 (-18%)
av2_cpu_v2	expl.	+10.5	+1.1 (11%)	+1.3 (12%)	+2.2 (21%)	-1.1 (-11%)
av2_gpu_full	conf.	+15.1	+2.3 (15%)	+1.8 (12%)	+4.1 (27%)	-1.0 (-6%)

B. Q2: Does covariate reweighting fix it?

No. Weighting the calibration scores by an estimated, label-free likelihood ratio increased the gap on every forward model by about two points (column "rew." in Table I) and left the reverse direction unchanged. Reweighting AV2 scenes to the nuScenes mix of speeds and manoeuvres even moves coverage the other way (Table II): nuScenes' scene mix is easier, so composition cannot explain the loss. This is the pattern expected when the conditional distribution of the future given the scene changes, which no reweighting on the scene can repair (Proposition 1).

C. Where does the loss come from?

The loss is concentrated in turns. On the confirmatory model, right-turn coverage falls by 41 points from AV2 to nuScenes, against 11.2 points for straight driving (Fig. 3). It is largest in Singapore, where traffic drives on the left: the right-turn drop there exceeds the drop in Boston by 27 points. What AV2-trained forecasters miss is the wide, across-traffic turn of left-hand traffic. Injecting one measured input difference at a time into AV2 test inputs reproduces only part of the gap (Table II): a 2 Hz history explains 15%, lane dropout 12%, and both together about a quarter. Excess annotation noise and map-point density were measured and found absent.

D. Q3: What can label-free methods do?

Little, and not reliably. The model-normalised score (Sec. III-A) reduces the AV2→nuScenes gap by 27% (to 11.0 points) but widens the reverse gap (to 8.6 points). On the reduced-budget models its effect ranges from a 49% reduction to a 52% increase. A domain-classifier shift monitor correlates with per-city coverage loss at Spearman $\rho \leq 0.47$ on every model, below the registered bar of 0.5. Adaptive conformal inference [13] converges only with a stream of labels; frozen after 1000 labels, its threshold covers $\geq 90\%$ of held-out scenes in only 40–50% of streams. These results agree with Proposition 1: without target labels there is no general fix.

E. Q4: How many labels to audit and to repair?

Three label budgets answer three different questions: how many labels to *detect* that the guarantee has failed, how many to return a threshold with a *certified* (training-conditional) guarantee, and how many before that threshold is also *small enough to be useful*. We report all three; the second does not imply the third.

Audit (detect). With $k = 1000$ labelled target agents the exact binomial audit (Sec. III-B) detects the gap with probability 1.00 on both confirmatory models, at a false-alarm rate of at most 7% on a shift-free split (Table I).

Repair with exchangeable labels (certify). When $k = 1000$ agents are labelled at random, certify-or-recalibrate (Sec. III-D) covers $\geq 90\%$ of held-out agents in 94–98% of label draws, against 50–60% for plain recalibration, at a region 1.18–1.23× the oracle's.

Repair with whole-scene labels (certify at the scene level). Real labels arrive scene by scene, which breaks agent-level exchangeability. We draw m labelled scenes from a pool disjoint from the evaluation scenes (nuScenes train for AV2→nuScenes, the Waymo calibration half for the Waymo targets) and evaluate the scene-averaged miscoverage $R(\hat{q})$ exactly on the evaluation pool, 1000 draws per m (Fig. 4). The scene-level rule with the betting p-value is valid at every m on all four transfers ($\Pr[R(\hat{q}) \leq \alpha] \geq 0.94$). The agent-level rule is not: on AV2→nuScenes, with a mean of 65 agents per scene (median 53), it is valid in 56% of draws at $m = 40$ and 4% at $m = 300$, and one-agent-per-scene split CP [37] in 57%. On Waymo and AV2, with few agents per scene, the agent-level rule stays valid in at least 65% of draws: agent-level guarantees fail where scenes are large.

Useful. Validity is not usefulness. With $m \leq 40$ scenes the scene-level rule abstains in every draw, correctly reporting that it cannot certify any finite threshold rather than returning an unsupported one. The abstention rate falls with m but is not negligible close to the recommended budget: on AV2→nuScenes it is 57% at $m = 80$ and still 6% at $m = 100$. It returns a usable region (at most twice the oracle area, abstaining in $\leq 5\%$ of draws) from $m = 150$ scenes for AV2→nuScenes, AV2→Waymo and nuScenes→Waymo, and from $m = 200$ for nuScenes→AV2, whose single-agent scenes carry less information each. At that budget the repaired AV2→nuScenes radius is 11.3 m against an oracle of 8.2 m. The betting p-value gives smaller regions than Hoeffding–Bentkus at large m (1.6× vs. 2.1× the oracle area at $m = 200$ on AV2→nuScenes).

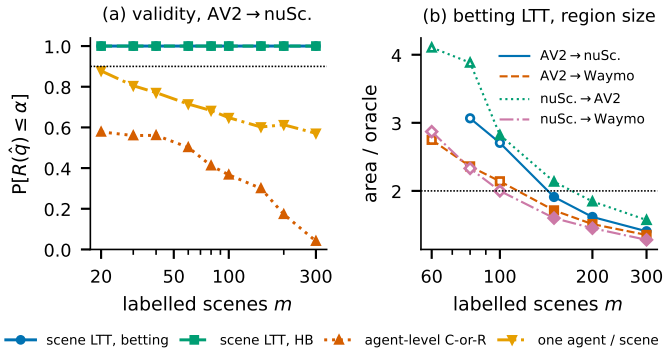


Fig. 4. Repair with m whole labelled scenes, disjoint label and evaluation pools, 1000 draws per m. (a) Probability that the repaired threshold controls scene-averaged miscoverage at α (dotted line: $1-\delta$). (b) Median region area relative to the oracle; hollow markers: threshold infinite in $> 5\%$ of draws.

F. Exploratory models

Three reduced-budget AV2 models and one reduced-budget nuScenes model (Table I) were analysed before the confirmatory models existed. Each reproduces the sign of every confirmatory finding: under-coverage on the target, no help from reweighting, losses concentrated in turns, and agent-level failure under scene clustering. The magnitudes vary by a factor of three, which is why the claims above rest on the confirmatory models alone.

VI. Discussion

A planner that trusts a source-calibrated region on a new dataset trusts a region that is too small. At $\alpha = 0.1$ the confirmatory models lose 15.1, 7.7, 23.4 and 18.3 points of coverage on the four transfers we tested, and the loss is silent: nothing observable without labels signals it. Three results bear directly on practice.

Label-free fixes are not safe defaults. Covariate reweighting made the gap larger on every forward model, and the normalised score helped in one direction and hurt in the other. A practitioner who applies either and stops can be worse off than one who does nothing.

Auditing is cheap; repairing is not. About 10^3 labelled target agents suffice to detect a gap of this size with high power. Repairing it with a guarantee that holds under scene clustering needs whole labelled scenes: roughly 150 to 200 of them before the region is both finite and within twice the oracle area. Below that budget the correct behaviour is to abstain, which the scene-level rule does.

The unit of labelling matters. Agent-level guarantees failed exactly where scenes are large (nuScenes, mean 65 and median 53 agents per scene) and held where they are small (Waymo, AV2). Any certification protocol for driving data should state its labelling unit and resample at that unit.

a) Mechanism: Two observations narrow the cause. The measured input differences explain about a quarter of the gap, and the loss concentrates in turns under left-hand traffic, which AV2 does not contain. Both point to a change in the conditional behaviour of road users rather than in the scenes themselves, consistent with Proposition 1. That the most accurate model shows the largest gap is consistent with a sharper, more confident source fit that extrapolates worse; testing this needs more architectures than we trained.

b) Limitations: (i) One architecture family; the conclusion that accuracy and coverage validity can diverge should be tested on others. (ii) The measured input differences leave most of the gap unexplained, and unmeasured causes cannot be ruled out. (iii) Coverage is marginal over agents or scenes, not conditional on a scenario; the manoeuvre breakdown is a diagnostic, not a guarantee. (iv) Proposition 1 covers scale shifts only. (v) The repair budget is measured on three datasets and may differ on others; the registered evaluation design (disjoint pools, exact evaluation) should be reused rather than the numbers.

VII. Conclusion

A conformal guarantee for trajectory prediction does not transfer across driving datasets. On three datasets and four transfer pairs, a forecaster calibrated at 90% covered between 67 and 82% of target futures, and a more accurate forecaster lost more coverage, not less. Reweighting does not repair the loss and no label-free method can in general. Labelled target data can: a scene-level Learn-then-Test rule with a correctly capped betting p-value restores the guarantee at every budget we tested and gives a usable region from about 150 labelled scenes. We recommend that any transferred conformal region be audited on target labels and, where it fails, recalibrated at the scene level before it is used for planning.

Data and Code Availability

Argoverse 2, nuScenes and Waymo Open Motion are third-party datasets available under their own licences from https://www.argoverse.org, https://www.nuscenes.org and https://waymo.com/open. Code, configuration files, the registered hypotheses, the experiment log and all aggregate result files (no raw third-party data) are at https://github.com/hwaqas160/conformal-coverage-transfer and archived under doi:10.5281/zenodo.23054413 (CC BY 4.0). Every number, table and figure is regenerated from the result files by `paper/make_results.py`.

Conflict of Interest

The authors declare no competing financial interests or personal relationships that could have influenced this work.

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The corresponding author used Anthropic Claude to edit language, draft and revise text, write and debug analysis code, run and monitor experiments, and generate figures and tables from computed results. The authors reviewed and edited all content and take full responsibility for it.

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